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ORIGINAL RESEARCH

Deep Learning-Infused Hybrid Security Model for Energy Optimization and Enhanced Security in Wireless Sensor Networks

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Abstract

Many wireless sensors are placed ad hoc to form a wireless sensor network (WSN), monitoring system, physical, and environmental conditions. Base stations and nodes constitute the system. The WSN's base station connects to the Internet, facilitating data sharing. These networks cooperatively transfer data to the base station while monitoring factors like sound, pressure, and temperature. The collected data undergo processing, analysis, storage, and mining. This study employs additional optimization and a deep learning approach to identify and isolate a rogue node from the busiest one based on various criteria. The deep learning model calculates probabilities using a sum-rule weighted method for request forwarding, reply forwarding, and data dropping. The planned task exhibits high throughput and reduced necessary time. Packet loss rates have decreased, with delay-related hyper metrics dropping from 70 to 42 ms. The percentage of missing packages has nearly threefold reduced from 23 to 8%. The adoption of deep learning eliminates hostile node behaviour, mitigating potential network failures.

Highlights

1. Enhanced Study optimizes with deep learning to pinpoint rogue nodes in the busiest, using sum-rule weighted probabilities for request and reply forwarding, and data dropping.
2. A deep learning model uses sum-rule weighting to compute probabilities for request handling, reply forwarding, and data dropping. This ensures high throughput and minimizes processing time in the planned tasks.
3. The proposed work achieves high throughput, reduced processing time, and a significantly lower packet loss rate, with nearly a threefold decrease in the percentage of missing packages.

Keywords WSN · Optimization · Deep learning approach · Connected dominating set (CDS)

Introduction

Wireless sensor networks are a specific kind of wireless network that are made up of a large number of mobile, self-contained, minuscule, and low-powered devices that are referred to as sensor nodes termed motes. These networks surely include a sizeable number of minuscule, geographically distributed embedded devices that are networked in order to conscientiously gather, process, and convey data to the operators, and it has handled the computing and processing capacities [1]. The nodes in a network are the individual computers that work together to build the network. The sensor node is a gadget that is wireless, has multiple purposes, and is energy efficient. Motes are put to use in a wide variety of different industrial applications. A network of sensor nodes collects data from the surrounding environment with the purpose of achieving certain application goals. Transceivers can be utilised to make it easier for moles and mice to communicate with one another [2]. It's possible that a wireless sensor network may have anywhere from hundreds to thousands of individual motes. In contrast to sensor networks, ad hoc networks will have a smaller number of nodes that are not organised in any way [4].

Transceivers are vital in wireless sensor networks, simplifying communication between nodes. These devices,

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combining transmitting and receiving functions, enable seamless bidirectional data exchange. Facilitating wireless connectivity, transceivers eliminate the need for physical cables, enhancing flexibility in node deployment. Their role in transmitting sensor data to central controllers and receiving instructions ensures real-time responsiveness. Energy-efficient design, standardized protocols, and coordination mechanisms make transceivers pivotal for reliable communication in diverse environments. Essentially, transceivers serve as the linchpin, enabling effective and dynamic information flow among sensor nodes, fundamental for the functionality and success of sensor networks. The data that is received by the WLAN Access Point is analysed by a piece of software known as Evaluation Software [5]. This is done so that the report can be sent to the users so that they can perform additional data processing on it, such as processing, analysis, storage, and mining. Power-restricted nodes that rely on batteries or energy harvesting are among the main features of a WSN, along with some mobility, heterogeneity, homogeneity, scalability to large-scale deployment, resistance to adverse environmental conditions, usability, and cross-layer optimization. The three primary issues with the conventional layered method are: Each layer in the conventional layered technique lacks complete knowledge since different tiers cannot interchange information. The entire network cannot be optimized using the conventional layered approach [7].

Motes, small wireless sensor nodes, find practical industrial applications in scenarios such as predictive maintenance. In factories, motes embedded in machinery collect real-time data on vibrations and temperature, facilitating early fault detection and preventing costly breakdowns. In agriculture, motes deployed across fields monitor soil moisture levels, optimizing irrigation. Integrating deep learning into WSNs enhances adaptability. For instance, in smart grids, deep learning algorithms analyze complex data patterns to predict energy demand and optimize distribution [8, 9]. In environmental monitoring, deep learning aids in recognizing abnormal patterns, providing early warnings for natural disasters. This integration ensures WSNs efficiently navigate dynamic sensor settings, delivering impactful solutions in challenging real-world environments. In order to enhance the transmission performance, including data rate, energy efficiency, quality of service (QoS), etc., the cross-layer can be employed to create the best modulation. It is possible to think of sensor nodes as little computers with very simple interfaces and parts. They typically include a processing unit with constrained memory and computational capabilities, sensors, or microelectromechanical systems (MEMS), a communication device (typically radio transceivers or alternatively optical), and a power source, typically in the form of a battery. Energy harvesting components, secondary application-specific integrated Circuits (ASICs), and perhaps secondary communication interfaces are further potential features [10].

Deep learning is being used in WSN applications for many different reasons [11]. As follows:

1) Because the sensors are dynamic, a model that can adapt to different settings and perform well in them is necessary.

2) The utilization of sensor networks may be employed to address problems that occur in any dangerous locations. Designers to control themselves and gather information in unforeseen situations must build deep learning models.

3) Access to enormous volumes of data is necessary, but sometimes it might be difficult to draw out the important relationships between the data [11].

Designing routing protocols for Wireless Sensor Networks (WSNs) involves challenges like energy efficiency, scalability, and adaptability to dynamic topologies, addressing the constraints of sensor nodes and ensuring reliable data delivery. The process of designing routing protocols for WSNs is quite difficult because of several features that set them apart from wireless infrastructure with fewer networks [12]. Different kinds of routing difficulties are present with WSNs. The assignment of a universal identifying system to many sensor nodes is about as difficult. Therefore, Internet Protocol (IP)-based protocols cannot be used by wireless sensor nodes. A certain base station needs the information that is detected from various sources. However, in typical communication networks, this does not take place [14]. Since multiple sensing nodes can generate comparable data when detecting, the data generated is typically highly redundant. In order to leverage such redundancy, routing protocols, available bandwidth, and energy must be used [15]. The bandwidth, transmission energy relations, storage, capacity, and onboard energy of wireless nodes are also limited. The number of the most recent routing protocols has been calculated to address routing issues within WSNs as a result of these differences [16].

The subsequent paragraphs will delineate the framework for the remaining sections of the paper. In the second section, a concise overview of the pertinent literature is presented, while the third section elucidates the methodology employed, along with the theoretical underpinnings of the utilised methods. Sect. "The Proposed Work:" encompasses the presentation of both the outcomes derived from the simulation as well as a comprehensive analysis thereof. The "key findings" section, situated towards the conclusion of the chapter, serves as a concise compilation of the most noteworthy discoveries.

Past Related Work

Several academics worked on the popular study topic of clustering in the sensor networks field, with a recent increase in the number of research findings. While extending network lifespan and enhancing network performance are all protocols' ultimate goals, each protocol focuses on increasing the grouping characteristics in a particular stage [18]. Here, we present a thorough analysis of cluster-based routing methods for homogenous networked sensors. They concentrated on the role that routing protocols played in the construction of clusters, CH selection, data aggregation, and data transfer phases of the clustering process [19].

WSNs have received a lot of attention in recent years and can be used in a variety of situations, both military and civilian [20]. It can be difficult to create routing protocols for WSNs that are efficient, reliable, and scalable. However, clustering routing techniques may generally match the limitations and difficulties faced by WSNs. As a result, it has been determined thus far that major efforts have been made in recent years to address the methodologies for designing efficient and effective clustered routing protocols for WSNs [21]. They provided a rather thorough overview of clustering routing techniques in WSNs in their paper. On the basis of quite specific clustering properties, we have also created a novel taxonomy of clustering routing algorithms for WSNs.

The WSNs have significantly increased in size in recent times, playing a significant role in the efficient selection and transmission of data. For networks, energy efficiency is a crucial concern, particularly for WSNs with their constrained battery capacities [23].

The authors of a study focused on the WSN-specific energy-efficient methods that have been created. They are divided into flat, hierarchical, query-based, coherent and incoherent-based, negotiation-based, location-based, mobile agent-based, multipath-based, and QoS-based categories. For a small network with fixed nodes, flat protocols can be the best option [24]. Due to link and processing costs, they become impractical in big networks. The hierarchical protocols try to address this issue and generate scalable, effective solutions. To effectively maintain the energy consumption of the sensor nodes, they segment the network into clusters, and they execute data fusion and aggregation to cut down on the volume of messages sent to the sink. Based on the sensors' energy reserves and their closeness to the cluster head, the clusters are formatted. Hierarchical protocols are therefore appropriate for sensor networks with high load and extensive coverage [25]. The Table 1 summarizes the similar work done of the same field.

Wireless Sensor Networks (WSNs) present unique challenges distinct from traditional wireless infrastructure, including resource constraints of sensor nodes, dynamic topology changes, and energy efficiency imperatives, demanding specialized routing protocols for effective communication in constrained and dynamic environments.

The Objective of the Work

Recommend a Deep Learning Based Security Model (DLBS) to optimize energy usage and enhance performance in wireless networks. Utilize supervised learning, integrate with existing security measures, and implement real-time adaptation for effective results.

The Proposed Work

In order to track any abnormalities in the network's nodes' behaviour, a deep learning model is created and deployed. The simulation was conducted using Network Simulator 3 (NS-3) version 3.35, which provides a comprehensive environment for modeling, simulating, and analyzing the performance of wireless sensor networks. The simulation setup involves deploying a wireless sensor network with nodes using a Deep Learning-Infused Hybrid Security Model. Each node employs energy optimization techniques alongside security protocols enhanced by deep learning algorithms. The network's performance is evaluated under varying conditions, measuring metrics such as energy consumption, throughput, latency, and security breach attempts to validate the model's effectiveness in energy optimization and enhanced security.

The datasets used include publicly available wireless sensor network data from the Intel Berkeley Research Lab and simulated data generated using NS-3. These datasets contain various sensor readings and network activities, providing comprehensive inputs for training and evaluating the deep learning model's performance in energy optimization and security enhancement.

The deep learning architecture employed in this research is a Convolutional Neural Network (CNN) integrated with a Long Short-Term Memory (LSTM) network, designed to optimize energy usage and enhance security in wireless sensor networks. The CNN layers are responsible for extracting spatial features from the sensor data, while the LSTM layers capture temporal dependencies and sequential patterns crucial for detecting security anomalies and optimizing energy consumption. The model consists of three convolutional layers with ReLU activation functions, followed by max-pooling layers to reduce dimensionality. These are connected to two LSTM layers, each with 128 units, to process the sequential data. Finally, a dense layer with a softmax activation function classifies

Table 1 The Existing Work done on the similar field

S. No	Citations	Research work	Findings
1	[3]	They provided a thorough analysis of several WSN energy management plans. These programs are divided into two groups: those that depend on energy provision and those that depend on energy consumption. Energy provision methodologies examine the properties of the energy source and create algorithms based on the energy accessibility to the sensor	The purpose of this survey was to evaluate the potential of various alternative energy sources and the many initiatives being made to utilize them effectively. When the supply and consumption of each node are taken into account, a network-wide energy-efficient protocol may more effectively control its operation
2	[6]	The proposition posits that a Wireless Sensor Network (WSN) represents a highly efficient monitoring methodology that can be effectively employed within the context of Public Administration (PA). This manuscript provides a comprehensive analysis and evaluation of the current advancements in Wireless Sensor Networks (WSNs). In order to accomplish the aforementioned objective, an initial step involved elucidating the diverse topologies of wireless sensor network (WSN) methodologies	Energy plays a crucial part in a WSN, hence the power sources used in WSN techniques, such as battery and solar power, were described. Second, the present
3	[9]	This work outlined the issue of energy holes in corona-based WSNs and discussed methods for addressing this issue. Analytical studies and approaches for balancing energy use were the primary focus of this review	Basic mathematical modelling of network connectivity and coverage, energy consideration, and the ideal width of coronas in the corona-based wireless sensor network (WSN) were also investigated in order to utilise these challenges in the development of solutions to the energy hole problem
4	[13]	This paper provides an overview of logical topologies and hierarchical routing. More specifically, there are five types of hierarchical topologies for WSNs	According to various logical topologies, many logical topologies for hierarchical WSNs have been analysed, including their features, benefits, and drawbacks. Furthermore, a number of hierarchical routing techniques for WSNs have been thoroughly examined
5	[17]	In this paper, authors have proposed a new an intelligent protocol called ELDC	It incorporates group-based protocol features that cause nodes to age more quickly than cluster-based nodes, extending their lifespan; Additionally, it uses ANN to train the protocol while taking into account parameters. This protocol differs from other state-of-the-art protocols, due to all of these aspects
6	[22, 26]	In this study, the authors presented a Genetic Fuzzy System (GFS) system as an alternative to our established routing algorithm for the purpose of performing clustering	The network lifetime is enhanced by adopting a two-phase CH selection method. Thus, as can be seen from the experimental findings, the network's overall energy consumption when routing packets is also significantly decreased

the outcomes. Hyperparameters used in training include a learning rate of 0.001, batch size of 64, and 50 epochs. The Adam optimizer is utilized for training, with categorical cross-entropy as the loss function. Dropout layers with a rate of 0.5 are employed to prevent overfitting, ensuring robust model performance.

Energy optimization is crucial for prolonging device lifespan and reducing costs. It enhances performance by ensuring efficient resource use. Before optimization, high energy consumption led to shorter device life and increased operational expenses.

The sum-rule weighted method is used to compute probabilities for different network actions such as request handling, reply forwarding, and data dropping. In this method, individual probabilities from multiple criteria or models are combined by summing them up, with each probability assigned a specific weight based on its importance. This weighted sum provides a balanced and comprehensive probability score, ensuring that the final decision takes into account various factors effectively. This approach enhances decision-making accuracy, optimizes network throughput, and minimizes processing time by prioritizing actions that contribute most significantly to overall network performance.

The deep neural network serves as the component vector in the encrypted wireless sensor network middleware. Logical inference, eigenvector analysis, and RL are its constituent parts. This concept incorporates flexibility in managing various scenarios using the network's underlying hidden layers. Certain features are applied as soon as the process of transmitting data begins. This is done by determining the harmful behaviour of the devices and the dynamic nature of the network, and then implementing these characteristics. These features include the characterization of the data loss and the data delaying process. The Fig. 1 illustrates the Connected Dominating Set founded Routing flow.

In this model, the first step is to create a cover set using the Cover set Detection Set technique, which is part of the Machine Learning approach. The proposed study emphasises a security model that can be utilised to detect anomalies in node behaviour during route formation, ensuring that all data is transmitted safely. The security model's algorithm is detailed below:

Security Model Algorithm:

1. CDS Establishment.

α—packet accelerating sum total

β—packet directed sum total

γ—packet droplet sum total

2. If $\sum_{i=0}^{n} N_i(\alpha, \beta, \gamma)$ (1)

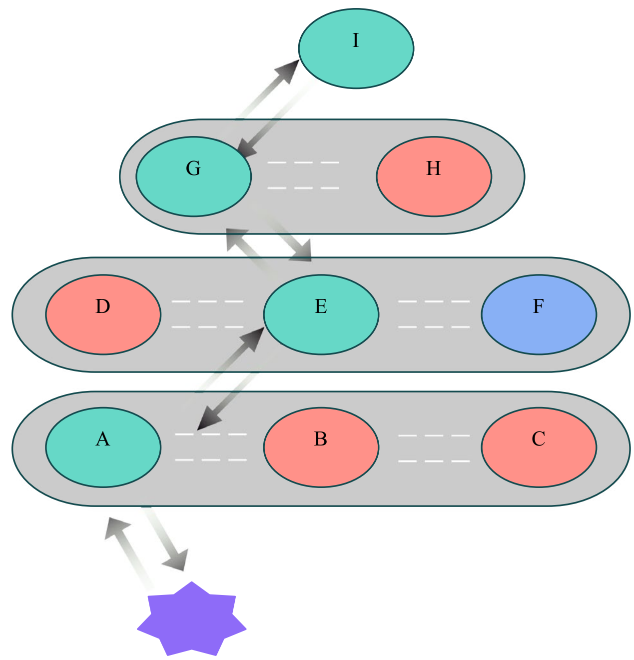

Fig. 1 Connected dominating set founded routing

3. Then $avg(\alpha, \beta, \gamma)$ is calculated.

4. Then $A = \rho(\alpha, \beta, \gamma)$ is computed.

5. Polynomial Kernal Function

$$\sum_{i=0}^{n}(\alpha, \beta, \gamma^T, a^d) \tag{2}$$

In this step, the parameters of the polynomial function are calculated.

In this case, "d" stands for the polynomial, represents the variable parameter.

6. you must calculate the Gaussian kernel function for the parameters.

$$K_1 = exp^{\frac{A_i}{\sigma^2}} \tag{3}$$

7. Neuron prototypical input- K_1

The goal of the detection system is to collect data to calculate metrics such the number of forwarded packets, sent packets, and dropped packets. Since deep learning's guiding premise is the system's collective intelligence, it could be used to supplement the process of behavioural distinction. Before that, regular patterns of behaviour are employed; these are calculated using the mean values of the aforementioned indicators [22]. It is possible to calculate the trained input parameter's variance by adding the absolute value of the differences that exist between individual values and the mean. The variance value is square-rooted to arrive at the standard deviation. Both the Gaussian kernel function and the linear polynomial kernel function are computed using the estimated parameters. This is how the good guys in the network are identified from the bad guys. These factors constitute the behaviour of the node and are taken into account while considering the input neurons. Neurons' primary function is to supply the data needed to calculate the aforementioned metrics, which includes the values of the metrics' probabilistic parameters. The hidden layer neuron count, the number of hidden layer neurons used, the number of hidden layer neurons generated, and the number of hidden layer neurons dropped are all represented by a set of four neurons. The transfer function is combined with a random value distribution to produce the initial weight matrix. Sigmoid and soft max activation functions are utilised when the state is in the bias position. Each neuron is compared to its corresponding neuron in the data set used for training.

The neuron error can be determined by entering the values into the appropriate spaces in the formula for the Euclidean distance. By carrying out the development of the subsequent neuron, we obtain the neuron with the smallest possible mistake. The function with minimal discontinuity is used to represent the continuous function for the parameter structure. The discontinuous function is used to build any transition between the nodes' typical and abnormal behaviour. In the activation procedure, the sigmoid function and the characteristics of linear perception are employed. The threshold value for the neuron's match is calculated at the beginning of each epoch for each hidden layer. Each successive layer receives the output of the preceding layer as its input.

The following input canonical partition, provided by Equation (below), is used to generate the hidden layer's probability dissemination.

$$Q = Sum(e^{-\in/KT}) \tag{4}$$

$\in$ – Dissemination Continual using Boltzmann,

K- Erudition factor and.

T: Knowledge Time Part.

Validating the two hypotheses, the alternate and the null, is how the behaviour categorisation of hypothetical circumstances is accomplished. The goal of these hypotheses is to distinguish between possible behavioural types based on a given set of characteristics.

Above, Fig. 2 depicts the architecture of the proposed deep learning-based model. Normal behaviour is based on the behaviour of the attackers, and the attackers' intent is based on the behaviour of the defenders. When malicious nodes are detected, the detector nodes send out a "attacker announcement" to the rest of the network, effectively cutting them off.

Result Analysis and Discussion

The proposed DLBS model is validated via an evaluation approach that takes into account a number of indicators. The following settings as shown in Table 2 are applied to the runtime environment:

Evaluation software in the context of a WLAN (Wireless Local Area Network) typically refers to software designed to assess and analyze the performance, efficiency, and overall functionality of the network. Overall, evaluation software plays a crucial role in maintaining and improving the performance of WLANs by providing insights into network behaviour and facilitating informed decision-making for network administrators. Metrics for system, service, and network performance have traditionally been linked.j Implementing the DLBS method allowed for a thorough review to be performed in terms of multiple classifications of metrics. Total energy used, remaining energy, and relative energy are all shown in the table, along with their fluctuation over time in seconds. The overall energy consumed by the network rises as the number of nodes grows. The quantity of energy that is required to keep DLBS operating has been cut back to some degree, which has resulted in an increase of 0.024% in the percentage of residual energy. To calculate the relative energy, divide the total residual energy by the greatest residual energy. By utilising a variety of assessment parameters, the performance of the suggested technique is evaluated and contrasted with that of the Optimised Energy Based Security Model (OEBS) and the Machine Learning Based Security Algorithm (MLBS).

PDR (Packet delivery ratio): PDR refers to the ratio of the number of packets received to the number of packets sent. The transfer of packets is the major factor contributing to the success of wireless networks. This delivery ratio is a success in terms of the PDR (product delivery ratio).

$$PDR = \frac{Recieved\ Packet\ Count}{Delivered\ Packet\ Count} \tag{5}$$

Throughput: The pace at which data packets are successfully conveyed from the sending node to the receiving node is referred to as the throughput of the network.

$$Throughput = \frac{Forwaded\ data}{Transmission\ time} \tag{6}$$

End-to-end delay: To a lesser extent End-to-end (E2E) delay has a number of advantages, two of which being decreased power consumption and increased reliability. Therefore, reducing the amount of time spent waiting will increase both the efficiency and dependability of the process. The E2E delay is a measurement used to determine how

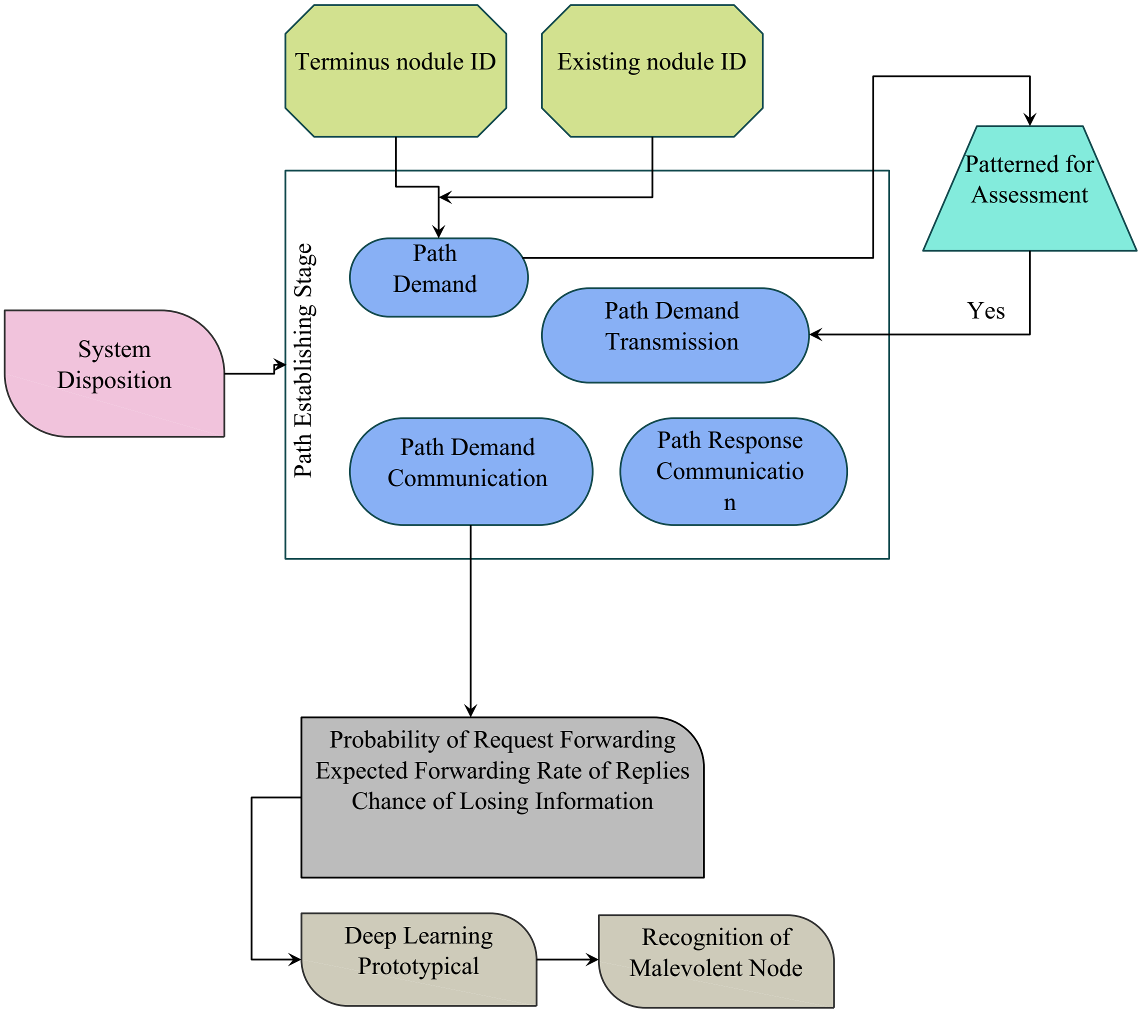

Fig. 2 DLBS model architecture

Table 2 Imitation arrangement

Constraint	Assessment
Nodes count	500 sensors
MAC category	Sensor Mac with IEEE802.11
File category	Precedence backlog
Exposure zone	80 m
Topology zone	500 × 500
Antenna category	Omnidirectional
Package extent	512
Construction category	UDP
Solicitation category	Instrument
Package generation interval	0.1–0.3
Statistics degree	3mbps
Imitation phase	20—100 s
Attacker count	5 DDOS
Primary energy	500 Joules

Table 3 Analysis of PDR (in %)

S. no	Intervals (s)	OBES	MLBS	DLBS
1	100	85.51	89.15	90.68
2	150	86.64	89.63	91.24
3	200	86.11	89.86	91.14
4	250	86.56	89.92	91.49
5	300	86.91	89.98	91.83

long it takes a packet to travel from one node to another. The delay from beginning to end takes into account the amount of time spent on activities such as the processing of data, transmission, and reception.

$$\begin{aligned} End\ to\ end\ Delay = {} & Time\ for\ (Data\ transmission \\ & + Data\ processing + Data\ delivery) \end{aligned} \tag{7}$$

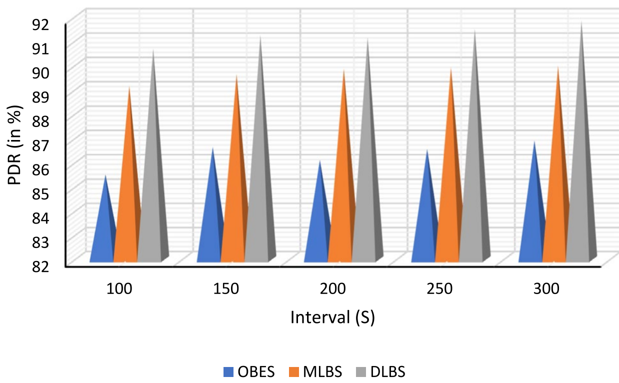

Fig. 3 Analysis of PDR (in %)

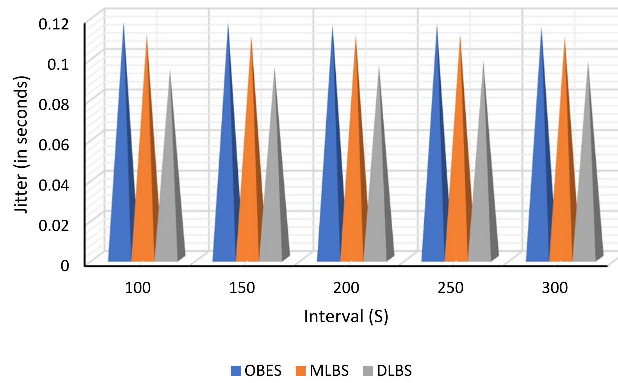

Fig. 5 Analysis of Jitter (in seconds)

Table 4 Analysis of delay (in seconds)

S. no	Intervals (s)	OBES	MLBS	DLBS
1	100	0.062	0.053	0.048
2	150	0.063	0.051	0.046
3	200	0.063	0.050	0.044
4	250	0.062	0.050	0.042
5	300	0.062	0.048	0.041

Table 5 Analysis of Jitter (in seconds)

S. no	Intervals (s)	OBES	MLBS	DLBS
1	100	0.117	0.111	0.094
2	150	0.117	0.110	0.095
3	200	0.116	0.111	0.096
4	250	0.116	0.111	0.098
5	300	0.115	0.110	0.098

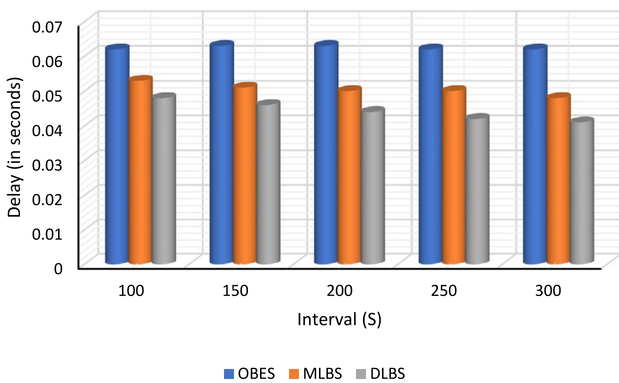

Fig. 4 Analysis of Delay (in seconds)

Table 6 Analysis of throughput

S. no	Intervals (s)	OBES	MLBS	DLBS
1	100	35.48	36.57	38.29
2	150	35.84	36.71	39.15
3	200	35.92	37.28	39.69
4	250	35.96	37.53	40.59
5	300	36.15	38.14	40.73

Table 3 compares the packet delivery ratio from the previous phase to that of a continuous growth in nodes. Figure 3 shows that the suggested DLBS model appears to utilise less energy, with savings approaching 50% as more nodes are added.

Table 4 compares the simulation phase with the preceding phase in terms of the network delay over a short time frame. Figure 4 shows that even if using Deep Learning is superior in benchmarking, the proposed model DLBS presents the lowest delay network as nodes get scaled up.

Experimental network's Jitter during one phase to that during the previous phase. Figure 5 shows that the suggested DLBS model has the least jitter compared to the other common network models (Table 5). Moreover, it is anticipated that the jitter will drop by somewhere in the neighbourhood of forty percent as the scale of the network grows in terms of the number of nodes.

The experimental network's throughput was measured over a short period of time and compared to other models in Table 6. Figure 6 shows that compared to the traditional network models, the suggested DLBS model has higher overall throughput. When the number of nodes in a network is increased, the throughput almost doubles.

In Table 7, we can see the average amount of energy used over a short period of time and how it compares to the previous stage. Figure 7 shows that as the number of nodes increases, the energy consumption of the proposed DLBS model decreases, eventually reaching a value of 52% lower.

Thus, the results of the suggested work have been demonstrated to be high throughput and a decrease in time. The percentage of lost packets has been reduced. The delay-related

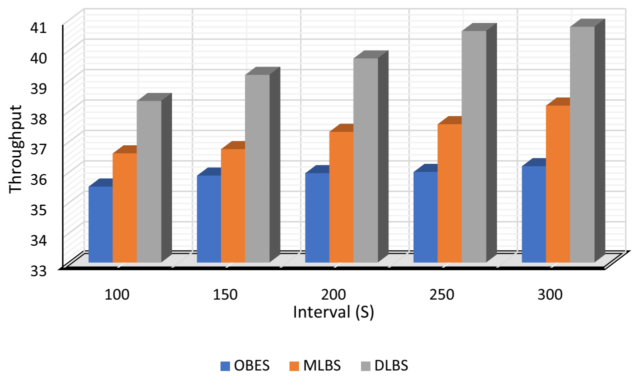

Fig. 6 Analysis of Throughput

Table 7 Analysis of average consumed energy (in J)

S. no	Intervals (s)	OBES	MLBS	DLBS
1	100	0.0381	0.0312	0.0319
2	150	0.063	0.0496	0.0553
3	200	0.0883	0.0672	0.0714
4	250	0.114	0.0841	0.0872
5	300	0.118	0.0852	0.0876

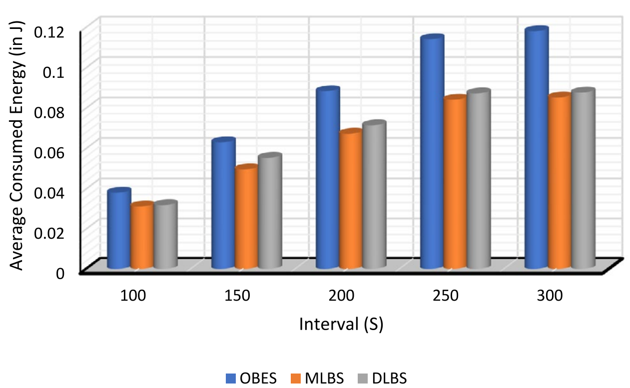

Fig. 7 Analysis of average consumed energy (in J)

hyper parameters are reduced from 70 to 42 ms. The percentage of lost packets has dropped from 23 to 8%, an almost threefold improvement. With the use of deep learning, malicious node conduct that causes the network to fail has been eliminated as a potential cause of failure.

Conclusion and Future Scope

The estimated key parameters that measure the QoS of any network are latency and lifetime. This is because there are many factors that degrade the QoS of any network. One common reason of both the estimations are the packet dropping ratio which is caused owing to DDoS attacks. Thus, the research is condensed into proposing a protocol that minimises the cost of maximising network lifetime while also minimising the cost of minimising packet dropping. The work is founded on the idea that it is possible to alleviate the issues created by malicious nodes, which are more difficult to isolate than in previous models, by maintaining the fittest node as active for normal communications and marking hyperactive nodes as malicious. This will allow for the problems caused by malicious nodes to be mitigated. A deep learning model for identifying the malicious node has been offered as a solution to these challenges. This model works by computing the probability of request forwarding, reply forwarding, and data dropping in a sum-rule weighted method.

High throughput and reduced time required are established properties of the proposed work. There has been a decrease in the overall rate of packet loss. There has been a drop from 70 to 42 ms in the delay-related hyper metrics. There has been a near threefold reduction in the percentage of missing packages, from 23 to 8%. The adoption of deep learning has removed hostile node behaviour that might bring down a network as a potential failure mode.

Several potential uses and approaches for the future could enhance the present process. Since transferring patients' private information on time and without error is crucial in the healthcare sector, the proposed method could be put into practice there. Future work in energy optimization aims to advance performance through innovative algorithms and adaptive strategies. Research will focus on refining existing optimization techniques and exploring cutting-edge approaches to achieve even greater efficiency gains in diverse applications and environments.

Authors Contribution Ramu K: Consumption and design of study, S. V. S. Rama Krishnam Raju: Acquisition of the Analysis and interpretation of the data, Satyanand Singh: Drafting, Venubabu Rachapudi: Formalization an editing, Anitha Mary M: Review and investigation**,** Vandana Roy: Conceptualization, Investigation, Shubham Joshi: Analysis.

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Data Availability Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval This article does not contain any studies with animals performed by any of the authors.

References

1. Jin Wang YU, Gao XY, Feng Li & Hye-Jin Kim 2018, "An enhanced PEGASIS algorithm with mobile sink support for wireless sensor networks", Wirel. Commun. mobile Comput., pp. 1–9.
2. Kai Arulkumaran, Marc Peter Deisenroth, Miles Brundage & Anil Anthony Bharath 2017 "A brief survey of deep reinforcement learning". arXiv:1708.05866
3. Khan F, Memon S & Jokhio SH 2016, "Support vector machine-based energy-aware routing in wireless sensor networks", in: 2016 2nd International Conference on Robotics and Artificial Intelligence (ICRAI), 01–02 Nov 2016, Rawalpindi, Pakistan, pp. 1–4.
4. Khan JA, Qureshi HK, Iqbal A. Energy management in wireless sensor networks: A survey. Comput Electr Eng. 2016;41:159–76.
5. Krishnan & Nagamuthu SS β019, "Denial of Service (DoS) Detection in Wireless Sensor Networks Applying Geometrically Varying Clusters", International Conference on Computer Networks and Communication Technologies, Springer, Singapore, pp.1023–1030.
6. Anisi MH, Abdul-Salaam G, Abdullah AH. A survey of wireless sensor network approaches and their energy consumption for monitoring farm fields in precision agriculture. Precision Agric. 2015;16(2):216–38.
7. Wazirali R, Ahmad R, Al-Amayreh A, Al-Madi M. Khalifeh, A, "Secure Watermarking Schemes and Their Approaches in the IoT Technology: An Overview." Electronics. 2021;10:1744.
8. Bouaziz M, Rachedi A. A survey on mobility management protocols in Wireless Sensor Networks based on 6LoWPAN technology. Comput Commun. 2016;74:3–15.
9. Asharioun H, Asadollahi H, Wan TC & Gharaei N, 2015,,,A survey on analytical modeling and mitigation techniques for the energy hole problem in corona-based wireless sensor network", Wireless Personal Communications, vol. 81, no. 1, pp. 161–187.
10. Al-Kashoash HAA, Kharrufa H, Al-Nidawi Y, Kemp AH. Congestion control in wireless sensor and 6LoWPAN networks: Toward the Internet of Things. Wirel Networks. 2019;25:4493–522.
11. Moridi MA, Kawamura Y, Sharifzadeh M, Chanda EK, Wagner M, Okawa H. Performance analysis of ZigBee network topologies for underground space monitoring and communication systems. Tunn Undergr Sp Technol. 2018;71:201–9.
12. Ertürk MA, Aydın MA, Büyükakkaşlar MT, Evirgen H. A Survey on LoRaWAN Architecture. Protocol and Technologies Futur Internet. 2019;11:216.
13. Liu X 2015,,,A typical hierarchical routing protocols for wireless sensor networks: A review, IEEE Sensors Journal, vol. 15, no. 10, pp. 5372–5383.
14. Darabkh, K.A.; El-Yabroudi, M.Z.; El-Mousa, A.H. BPA-CRP: A Balanced Power-Aware Clustering and Routing Protocol for Wireless Sensor Networks; Elsevier: Amsterdam, The Netherlands, 2019; Volume 82, ISBN 9626530081.
15. Sah DK, Amgoth T. Parametric survey on cross-layer designs for wireless sensor networks. Comput Sci Rev. 2018;27:112–34.
16. Khashan OA, Ahmad R, Khafajah NM. An automated lightweight encryption scheme for secure and energy-efficient communication in wireless sensor networks. Ad Hoc Netw. 2021;115: 102448.
17. Mehmood Z, Lv J, Lloret εε & Umar, 2017, ,,EδDC: An artificial neural network based energy-efficient and robust routing scheme for pollution monitoring in WSNs", IEEE Trans. Emerg. Top Comput., no. 99 pp. 1–8.
18. Parmod Kumar, Anupam Baliyan, K. Ramalingeswara Prasad, N. Sreekanth, Parag Jawarkar, Vandana Roy, Enoch Tetteh Amoatey, "Machine Learning Enabled Techniques for Protecting Wireless Sensor Networks by Estimating Attack Prevalence and Device Deployment Strategy for 5G Networks", Wireless Communications and Mobile Computing, vol. 2022, Article ID 5713092, 15 pages, 2022. https://doi.org/10.1155/2022/5713092
19. V. Roy et al., "Network Physical Address Based Encryption Technique Using Digital Logic", International Journal of Scientific & Technology Research, Vol. 9, No. 4, 2020, Pp no.- 3119–3122.
20. Rana M, Mamun Q, Islam R. Lightweight cryptography in IoT networks: A survey. Futur Gener Comput Syst. 2022;129:77–89.
21. Sharma H, Haque A, Blaabjerg F. Machine learning in wireless sensor networks for smart cities: A survey. Electronics. 2021;10:1012.
22. Roy V. An Effective FOG Computing Based Distributed Forecasting of Cyber-Attacks in Internet of Things. Journal of Cybersecurity and Information Management. 2023;12(2):8–17.
23. Parmod Kumar, Anupam Baliyan, K. Ramalingeswara Prasad, N. Sreekanth, Parag Jawarkar, Vandana Roy, Enoch Tetteh Amoatey, "Machine Learning Enabled Techniques for Protecting Wireless Sensor Networks by Estimating Attack Prevalence and Device Deployment Strategy for 5G Networks", Wireless Communications and Mobile Computing, vol. 2022, Article ID 5713092, 15 pages, 2022. https://doi.org/10.1155/2022/5713092.
24. Arfat Ahmad Khan, Khalid K. Almuzaini, Víctor Daniel Jiménez Macedo, Stephen Ojo, Vinodh Kumar Minchula, Vandana Roy, MaReSPS for energy efficient spectral precoding technique in large scale MIMO-OFDM, Physical Communication, Volume 58, 2023, 102057, ISSN 1874–4907,
25. Cui L, Yang S, Chen F, Ming Z, Lu N, Qin J. A survey on the application of machine learning for Internet of Things. Int J Mach Learn Cybern. 2018;9:1399–417.
26. Arfat Ahmad Khan, Khalid K. Almuzaini, Víctor Daniel Jiménez Macedo, Stephen Ojo, Vinodh Kumar Minchula, Vandana Roy, MaReSPS for energy efficient spectral precoding technique in large scale MIMO-OFDM, Physical Communication, Volume 58, 2023, 102057, ISSN 1874–4907, https://doi.org/10.1016/j.phycom.2023.102057.

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